The Difficulty of Marathon

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

If you don't want to read the background of the problem, please skip directly to the last paragraph.

In 490 BC, after the Greek victory in the Battle of Marathon, the messenger Pheidippides ran approximately 40 kilometers from the Marathon Plain to Athens to announce the triumph, and died shortly after declaring victory. To commemorate this event, the marathon was introduced as an event at the first modern Olympic Games in Athens in 1896, with an initial distance of about 40 kilometers. The route was adjusted to 42.195 kilometers at the 1908 London Olympic Games, and this standard was officially adopted by World Athletics in 1921. The women's marathon became an official Olympic event at the 1984 Los Angeles Olympic Games.

Marathon can easily lead to extreme physical exhaustion. Runners often experience "hitting the wall" (glycogen depletion) between 30 and 35 kilometers, causing a breakdown of both physical strength and willpower. Long-distance running also poses injury risks to the knees, ankles, calves, and other body parts. Additionally, it is difficult to control the pace at the start; the second half of the race is a lonely mental struggle; weather changes affect physical comfort; improper timing of refueling can cause gastrointestinal discomfort and electrolyte imbalance. Participating in a marathon also requires high standards of daily systematic endurance and strength training. Participating without proper preparation carries a high risk of injury.

However, the biggest difficulty of marathon lies in the extremely low registration success rate. The success rate for popular marathons is often below 10%, and for beginners, it can be as low as 2%.

If you are a marathon beginner, what do you think is the biggest difficulty of running a marathon?

For each draw factor N and registration number M, you are considered successful if N divides M. Given the registration number and draw factor, determine whether you are selected.

Input

The first line contains a positive integer $T\,(1 \le T \le 100)$, representing the number of test cases.

The next T lines each contain two positive integers $M, N\,(1 \le M, N \le 100000)$, representing your registration number M and draw factor N.

Output

For each test case, output one line with an integer: output `1` if selected, and `0` if not selected.

Example

standard input	standard output
2 2026 2026 2 2025	1 0